

Introduction to Theory of Computation Final Exam

2022 Spring Term

Time Limit: 300 minutes.

Total number of problems: 8.

Total credits: 110. But if you get more than 100, then your final credits will only be 100.

- Some problems are easier and others are harder, and they are not necessarily arranged from easy to hard.
- You can directly use all the results proved in class or in your homework. Otherwise, please prove your claims.
- Organize your work, in a reasonably neat and coherent way. Try to be concise and accurate in your argument. Lengthy paragraphs that are hard to understand may receive very little credit.
- Mysterious or unsupported answers will not receive full credit. An incorrect answer supported by substantially correct explanations might still receive partial credit.

1. (20 points in total, 2 points for each) Decide if the following statements are TRUE (T) or FALSE (F) or UNKNOWN (U). You do not have to write down proofs for your answers.
- (a) () A language is not regular if and only if no regular expression can describe it.
 - (b) () Let $L = \{a^m b^n \mid m \neq n^3 + 1\}$. L is a regular language.
 - (c) () The union of a regular language and a context-free language must be context-free.
 - (d) () The class of non-context-free languages is closed under intersection.
 - (e) () Let $L = \{0^a 1^b 2^a \mid a, b \geq 0 \text{ and } a \leq b \leq 3a\}$. L is a context-free language.
 - (f) () There exists a language A such that both A and $\bar{A}$ ($\bar{A}$ is the complement of A) are not Turing-recognizable.
 - (g) () If $\text{NP} = \text{coNP}$, then $\text{BPP} \subseteq \text{NP}$.
 - (h) () $\text{P/poly} \subseteq \text{NEXP}$.
 - (i) () In the definition of IP, the prover is not allowed to be uncomputable functions.
 - (j) () If one-way function exists, then there exists a language $L \in \text{NP} \setminus \text{P}$ that is not NP-complete.

2. (10 points) Let $X = \{\langle M, w \rangle \mid M \text{ is a single tape TM that never modifies the portion of the tape that contains the input } w.\}$ Is X decidable? Prove your answer.

3. (14 points) Consider the following complexity class NAC^0 :

A language L is in NAC^0 if there exists a polynomial $p : \mathbb{N} \rightarrow \mathbb{N}$ and a uniform AC^0 circuit family $\{C_n\}$ such that for every $x \in \{0, 1\}^*$,

$$x \in L \Leftrightarrow \exists u \in \{0, 1\}^{p(|x|)} \text{ s.t. } C_{|x|}(x, u) = 1.$$

Which ones of the following assertions are true?

- (a) $\text{NAC}^0 \subseteq \text{NP}$;
- (b) $\text{NAC}^0 \subsetneq \text{NP}$;
- (c) $\text{NAC}^0 \not\subseteq \text{NP}$.
- (d) $\text{NP} \subseteq \text{NAC}^0$;
- (e) $\text{NP} \subsetneq \text{NAC}^0$;
- (f) $\text{NP} \not\subseteq \text{NAC}^0$.

Prove your answer.

4. (12 points) Prove that TQBF is complete for PSPACE under logspace reductions.

5. (12 points) Suppose that there is a randomized Turing Machine M_R that decides L with two-sided error probability at most ϵ . Prove that for every $n \in \mathbb{N}$, there exists a small set S of size $2^{O(\log(n) + \log(1/\delta))}$ such that

$$\forall x \in \{0, 1\}^n, \Pr_{R \sim S} [M_R(x) \neq L(x)] \leq \epsilon + \delta,$$

where $R \sim S$ means that R is randomly and uniformly sampled from the set S .

6. (14 points) Is it true that $NL \subseteq (\text{log-space uniform}) NC$? Prove your answer.

7. (15 points) Set Lower bound with perfect completeness.

Let $S \subseteq \{0,1\}^m$ be a set such that membership in S can be verified in time $\text{poly}(m)$. Let K be an integer and let k be such that $2^{k-2} < K \leq 2^{k-1}$.

Show that there exists an $AM[2]$ protocol such that, given K to both parties,

- if $|S| \geq K$, then the verifier accepts with probability 1;
- if $|S| \leq K/k^2$, then the verifier rejects with probability $\geq 2/3$.

(Hint: there may be multiple ways to solve this problem but you can consider the following points.

We can have the prover choose the hash function. It can choose several hash functions $h_1, h_2, \dots, h_t$, where $h_i : \{0,1\}^m \rightarrow \{0,1\}^k, i \in [t]$. Does there exist a sequence of hash functions such that $\cup_{i \in [t]} h_i(S) = \{0,1\}^k, t = O(k)$, and they are easy to be described and easy to be computed by the verifier?)

8. (13 points) (The skipped proof in showing OWF implies PRG.)

Show the following claim.

For any random variable $Y = (Y_1, Y_2, \dots, Y_n)$ over $\{0, 1\}^n$, for every $\epsilon > 0$, if there exists there exists a probabilistic polynomial time TM M s.t.

$$|\Pr[M(Y) = 1] - \Pr[M(Y_{[n-1]} \circ U_1) = 1]| \geq \epsilon,$$

then there exists a probabilistic polynomial time TM M' s.t.

$$\Pr[M'(Y_{[n-1]}) = Y_n] \geq 1/2 + \Theta(\epsilon).$$

Here $Y_{[n-1]} = (Y_1, Y_2, \dots, Y_{n-1})$, i.e. the first $n - 1$ coordinates of Y . U_1 is a one bit uniform random variable. $Y_{[n-1]} \circ U_1 = (Y_1, Y_2, \dots, Y_{n-1}, U_1)$.